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Project 1 – Buck Converter
2/23/2026

Constraints:

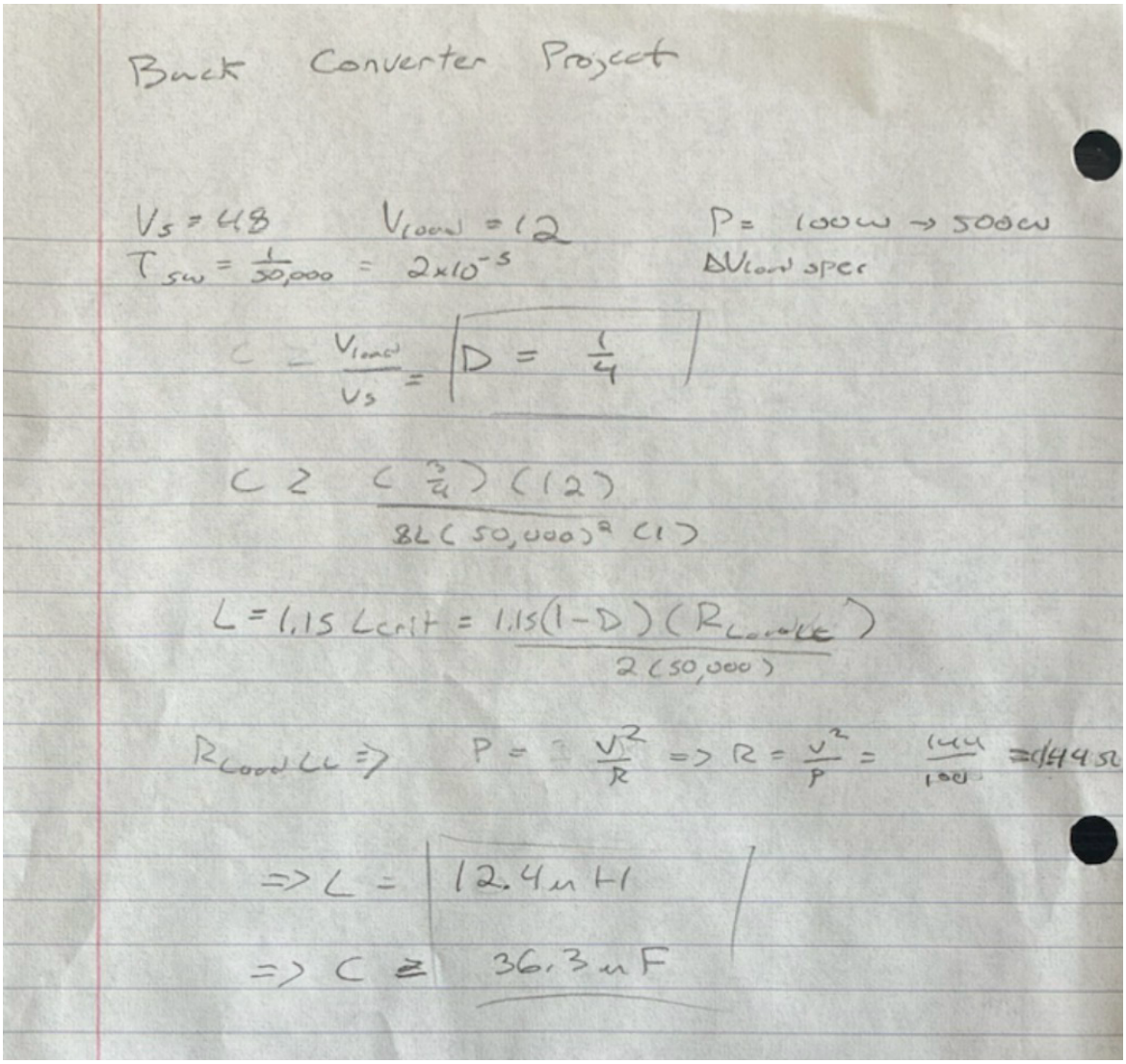
Design a Buck Converter circuit with an output voltage of 12V and maximum peak to peak ripple voltage of 1V, for a load range of 100W to 500W. A switching frequency of 50kHz and an Inductance of 1.15x Critical Inductance.

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1a.
Chosen Capacitor: 36.3 uF
Chosen Inductor: 12.4 uH
Heavy Load: R = 0.288Ω (500W)
Light Load: R = 1.44Ω (100W)

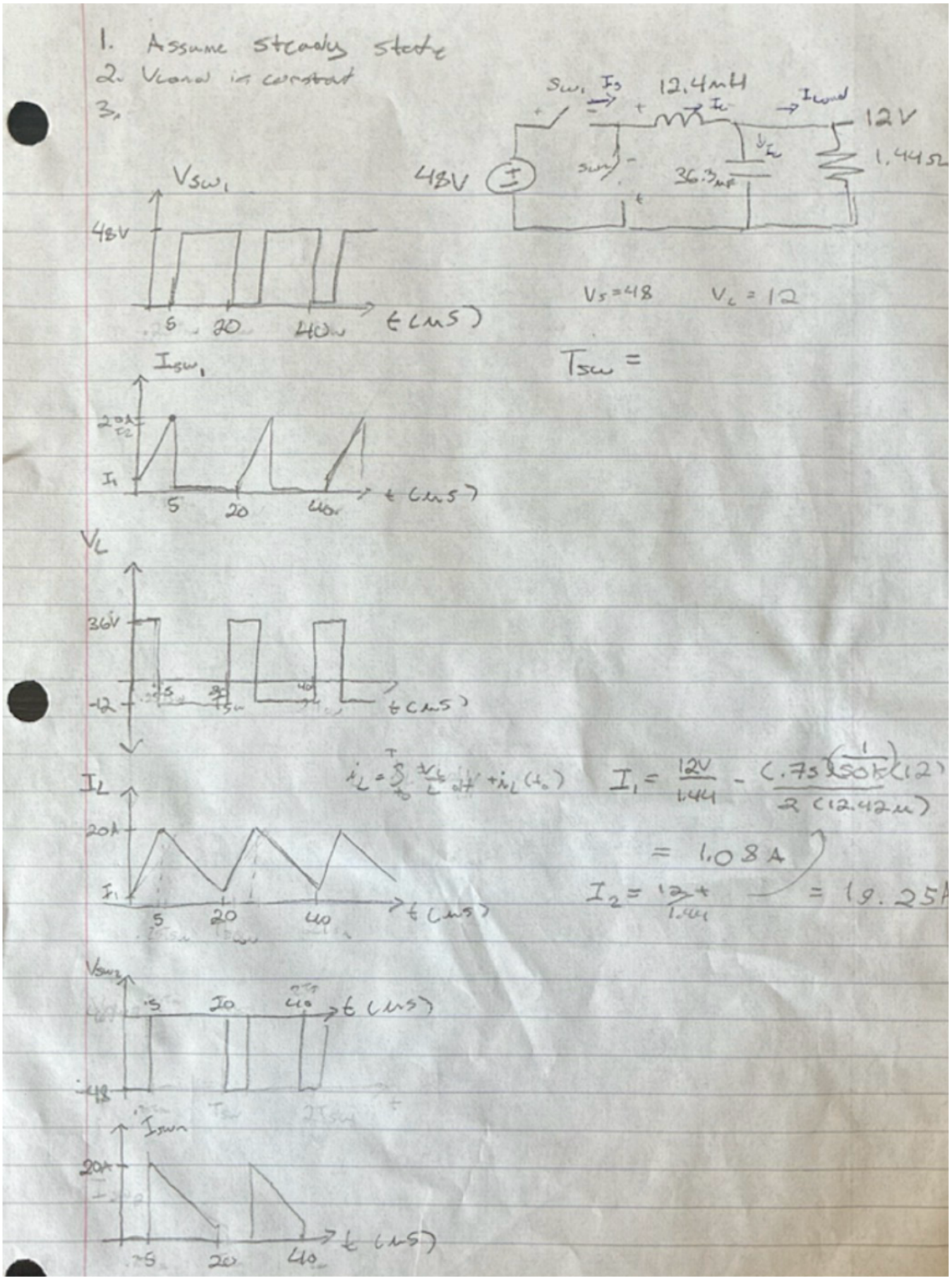
All equations were derived in class.



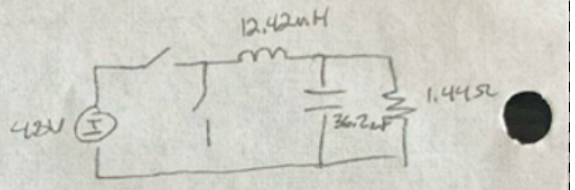
1b.

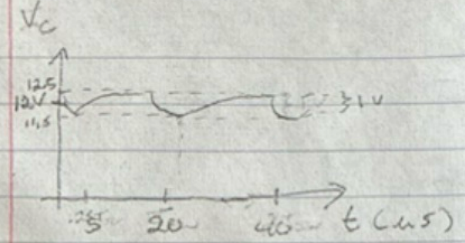
Determined Efficiency: 100%

All equations/graphs were derived in class.



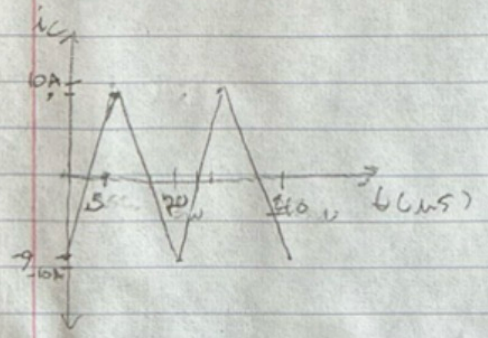
4. Recall assumption V_{load} constant





$$V_C = \frac{1}{C}\int I_C \frac{dV}{dt} = 27624 \int I_C$$

$$\Delta V = \frac{.75(12)}{8(12.42E-6)(50{,}000)^2 \cdot (36.2E-6)}$$



$$i_C = -\langle i_L \rangle + i_L(t)$$

$$\langle i_L \rangle = \frac{I_2 - I_1}{2} = 19.08\,A$$

$I_2 = 19.2$

$I_1 = 1.1$

$$\text{Expected Efficiency} = \frac{\langle \text{Power out} \rangle}{\langle \text{Power in} \rangle}$$

$$= \frac{\langle I_R \rangle \cdot \langle V_R \rangle}{\langle I_{sw} \rangle \cdot 48}$$

$$\langle I_R \rangle = \langle I_L \rangle - \langle I_C \rangle = \langle I_L \rangle = 10.16A$$

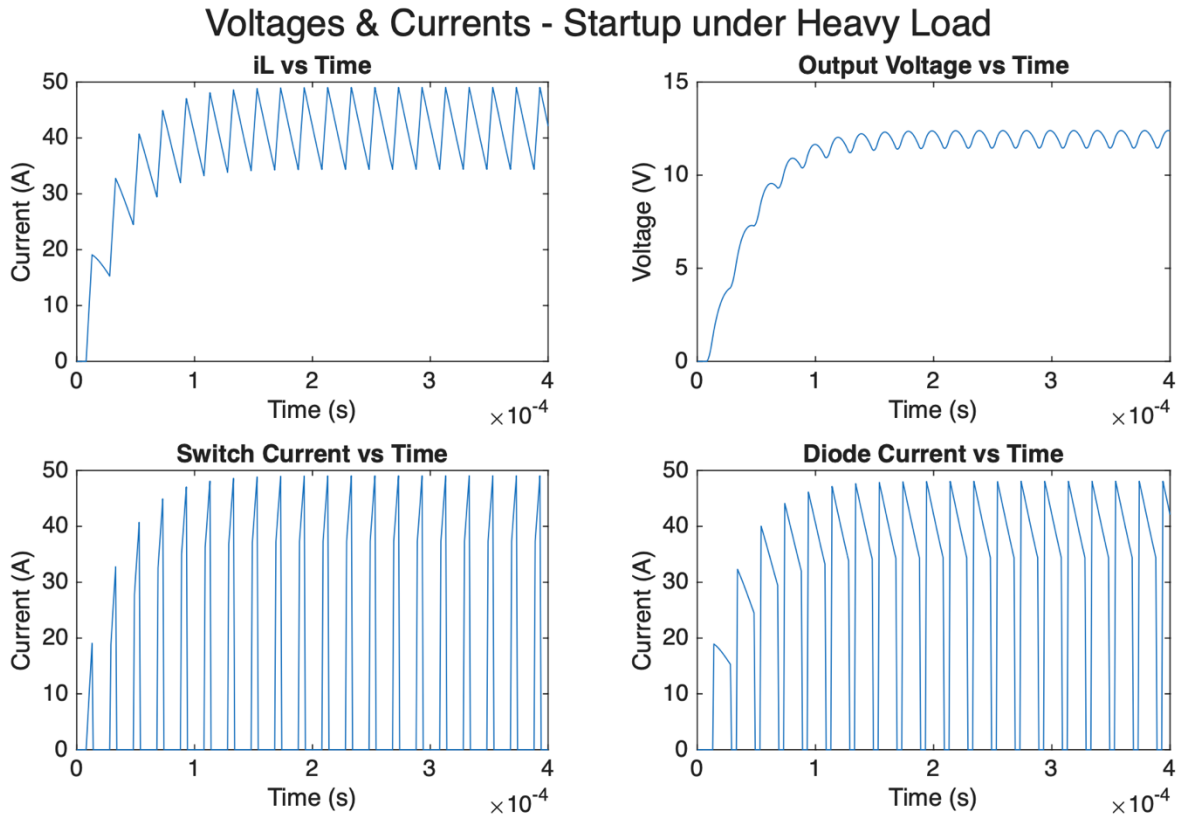
$$\langle I_{sw_1} \rangle = \frac{1}{T}\int_{t_0}^{5} I_{sw}\,dt = 50{,}000 \left(\int_0^{5\times10^{-6}} (I_1 + (I_2 - I_1)\frac{t}{5}) \right.$$

$$= 2.541\,A$$

$$\langle V_R \rangle = \langle V_C \rangle = 12$$

$$\Rightarrow \text{Eff} = \frac{12V(10.16A)}{48(2.541)} = 100\%$$

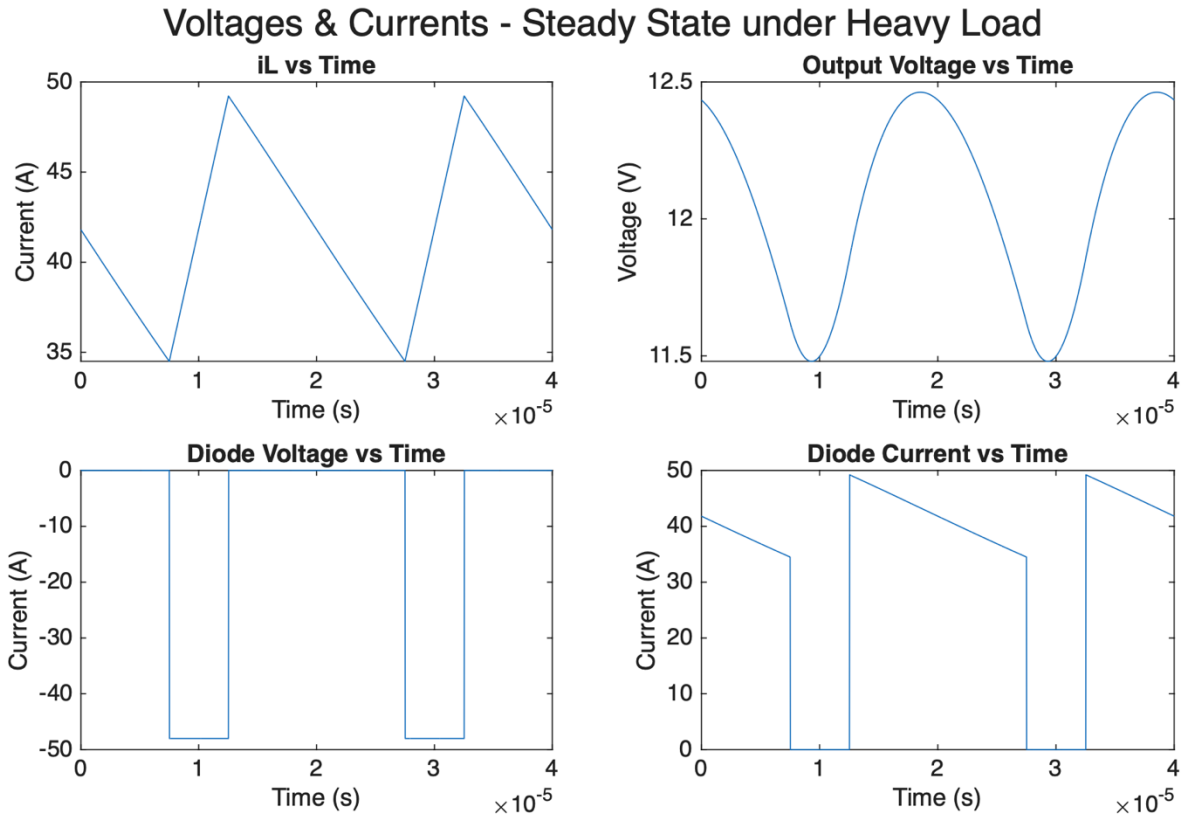
1di.



The average value for Load Current is 41.7500A.
The average value for Load Voltage is 12.0240V.
The average value for Switch1 is 10.4706A.

From these values, it can be determined that the input power is 502.6W (41.75A * 12.024V), and the output power is 502W (48V * 10.4706A). Both values are withing 1% of the expected 500W, and the expected efficiency from these values is 99.89%.

1dii.

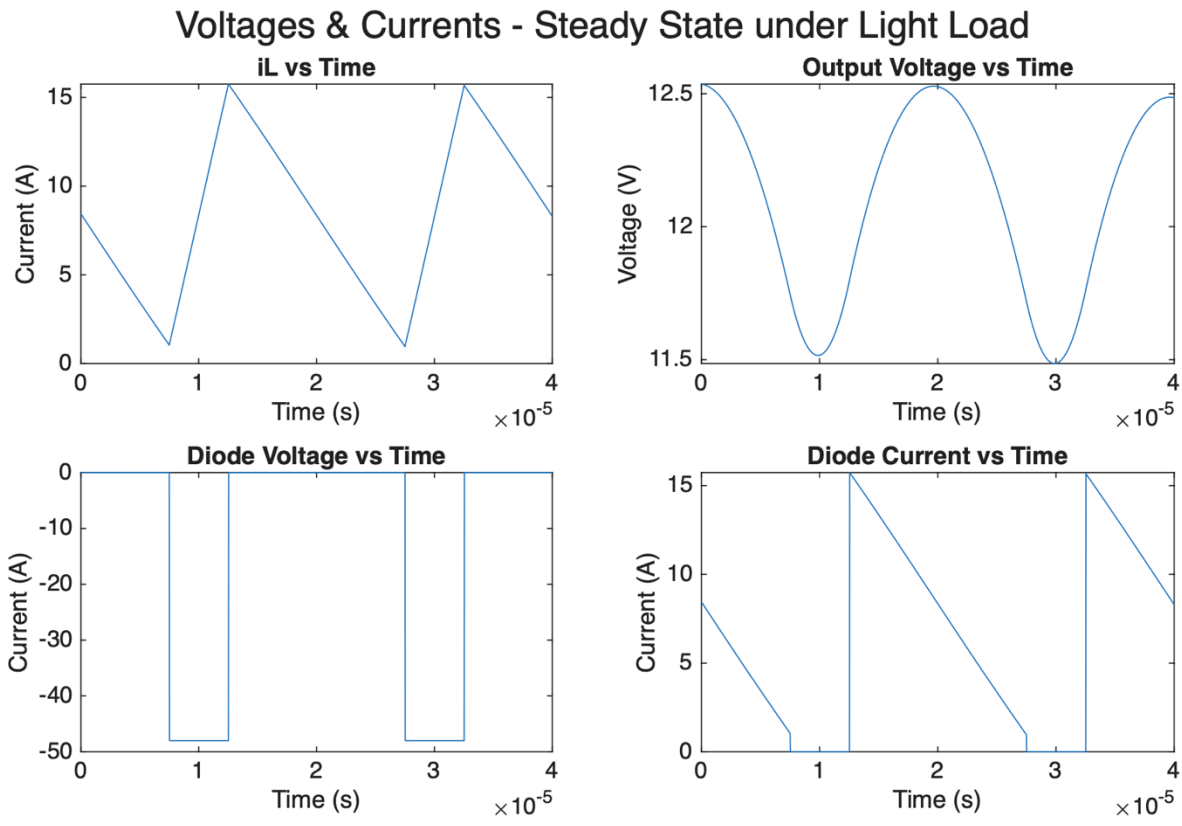


The average value for the Load Current for waveform is 41.75A.
The average value for the Load Voltage is 12.024V.
The average value for the Switch1 Current is 10.4706A.

From these values, it can be determined that the average input power is 502.56W (41.75A * 12.024V), and the average output power is 502W (48V * 10.4706A). Again, giving an expected efficiency of 99.89%. Given it is 2W off for both, this inconsistency can most likely be explained by an inaccuracy (rounding) in either the determined capacitance or inductance value. However, after increasing the accuracy of the capacitor, the values remained the same, I still believe the inaccuracy is due to some rounding error.

The ripple oscillates between 12.5 and 11.5 so it meets spec of 1Vpp maximum ripple around 12VDC.

1diii.



The average value for the Load Current for waveform is 8.3500A.
The average value for the Load Voltage is 12.0239V.
The average value for the Switch1 Current is 2.0973A.

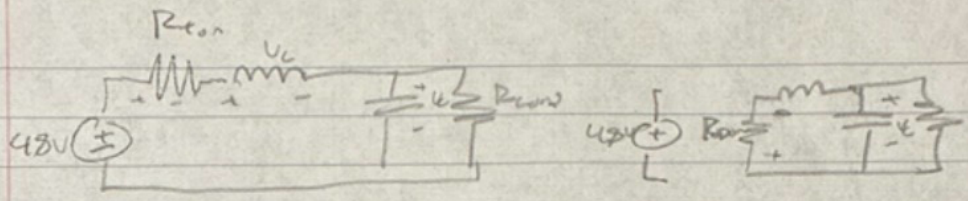
From these values, it can be determined that the average input power is 100.4W (8.35A * 12.024V), and the average output power is 100.67W (12V * 2.0973A). This time, giving an expected efficiency of 99.73%. The ripple oscillates between 12.5 and 11.5 so it meets spec of 1Vpp, as well as output voltage as it oscillates around 12.02 as the average value shows.

Based on the ripple, average voltage, and efficiency this ideal buck converter design can be deemed successful as it meets the spec.

1e.

Determined Duty Cycle: 0.2586 or 25.86%
Determined Currents: I1 = 34.246A, I2 = 49.053A
Expected Efficiency: 0.967 or 96.7%

The following equations were derived in hw3 for this class, and they have been copied down on the handwritten notes above. As a resistance is added to the diode in a non-ideal case, there is a voltage loss for both states. Once accounted for, the proper duty cycle can be calculated, and it will be higher than the original “ideal” duty cycle. Furthermore, the currents no longer drop from I1 to I2 linearly as they are based on VL which itself changes. This non constant drop is accounted for in the notes above.



$$(48 - \langle I_L \rangle R_{Ton} - \langle V_L \rangle - V_{Load})DT + (+R_{Ton}\langle I_L \rangle - \langle V_L \rangle - 12)\cdot(1-D) = 0$$

$\langle V_L \rangle = 0$

$$\Rightarrow 0 = (36 - \langle I_L \rangle R_{Ton})D + (-R_{Don}\langle I_L \rangle - 12) + (R_{Don}\langle I_L \rangle + 12)D$$

$R_{Ton} = R_{Don}$

$$R_{Don}\langle I_L \rangle + 12 = (48 + 0)D - 0.01)D$$

$$D = \frac{R_{Don}\langle I_L \rangle + 12}{48} = \frac{0.01\left(\frac{12}{0.288}\right) + 12}{48}$$

$$D = \boxed{0.2586}$$

$$\frac{I_1 + I_2}{2} = 41.67\ A$$

$$i_L(t) = \frac{V_{in} - V_{load}}{R_{Ton}}\left(1 - e^{-DT_{sw}/\tau}\right) + I_1 e^{-DT_{sw}/\tau}$$

$$\tau = \frac{L}{R_{On}}$$

$$\tau < D \Rightarrow$$

$$I_2 = \frac{V_{in} - V_{load}}{R_{Ton}}\left(1 - e^{-\frac{R_{Ton}}{L}DT_{sw}}\right) + I_1 e^{-\frac{R_{Ton}}{L}DT_{sw}}$$

$$+ > D \Rightarrow 1$$

$$I_1 = -\frac{V_{load}}{R_{Don}}\left(1 - e^{-\frac{R_{don}}{L}(1-D)T_{sw}}\right) + I_2 e^{-\frac{R_{don}}{L}(1-D)T_{sw}}$$

$$\begin{bmatrix} e^{-\frac{R_{Ton}}{L}DT_{sw}} & 1 \\ 1 & e^{-\frac{R_{Don}}{L}(1-D)T_{sw}} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} \frac{V_{in} - V_{load}}{R_{Ton}}\left(1 - e^{-\frac{R_{Ton}}{L}DT_{sw}}\right) \\ -\frac{V_{load}}{R_{don}}\left(1 - e^{-\frac{R_{don}}{L}(1-D)T_{sw}}\right) \end{bmatrix}$$

$$\Rightarrow I_2 = \frac{48-12}{0.01}\left(1 - e^{-\frac{(0.01)(0.259)\left(\frac{1}{30,000}\right)}{12.42\text{mH}}}\right) + I_1 e^{-\frac{0.01(0.259)\left(\frac{1}{30,000}\right)}{12.42\text{mH}}}$$

$$e^{\cdots} = 0.9958$$

$$\Rightarrow I_2 = 3600(1 - .9958) + I_1(0.9958)$$

$$= 14.98 + .9958 I_1$$

$$\frac{I_1 + I_2}{2} = \frac{12}{0.288} \Rightarrow I_1 + I_2 = 83.3$$

$$\Rightarrow I_1 + 14.98 + .9958 I_1 = 83.3$$

$$\boxed{I_1 = 34.246\text{A}}$$

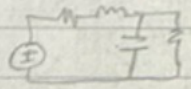
$$\boxed{I_2 = 49.053\text{A}}$$

~~$P_{in} = 48(0.259) =$~~

$$i_L(k+1) + \frac{\Delta t\, i_L(k+1) R_{Ton}}{L} = i_L(k) + \frac{\Delta t\,(sw_1 V_{in}(k+1))}{L}$$

$$\Rightarrow i_L(k+1) = \frac{i_L(k) + \Delta t\,(sw_1 V_{in}(k+1))}{1 + \Delta t\, R_{Ton}/L}$$

Efficiency



or ϕ

$\Rightarrow$ Assuming straight line

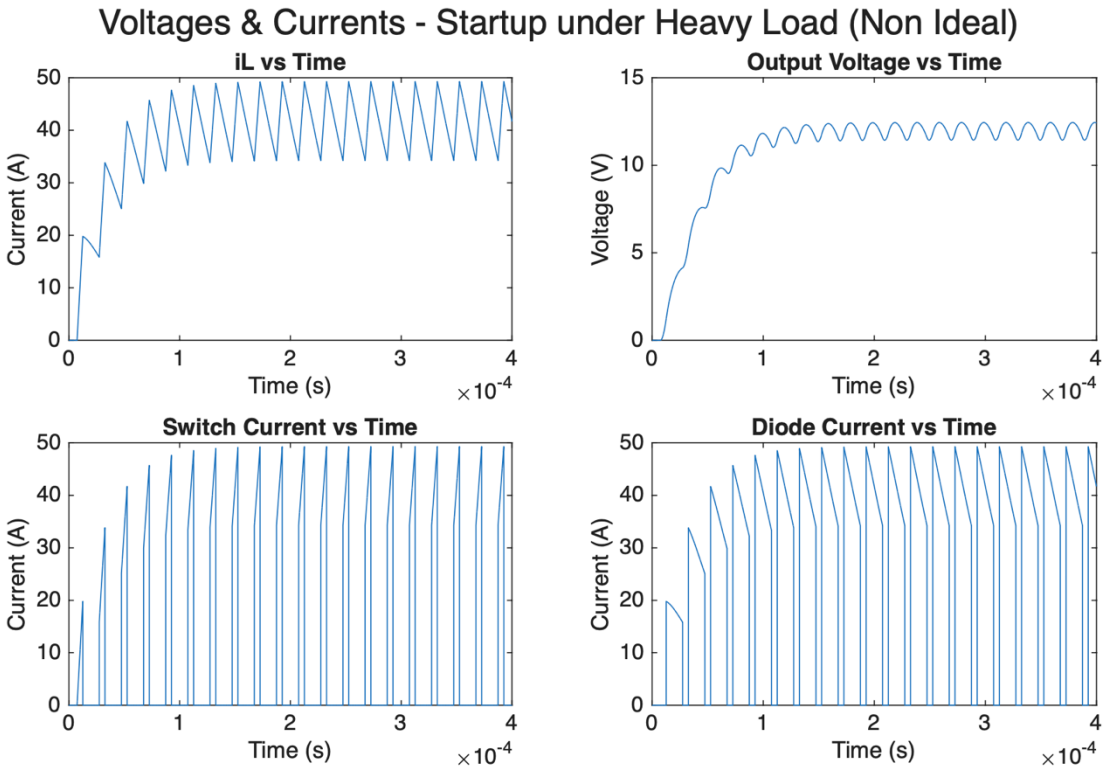
$$\langle I_L \rangle \cdot V_S \cdot 0.259 = P_{in} = 48(41.655)(.259)$$

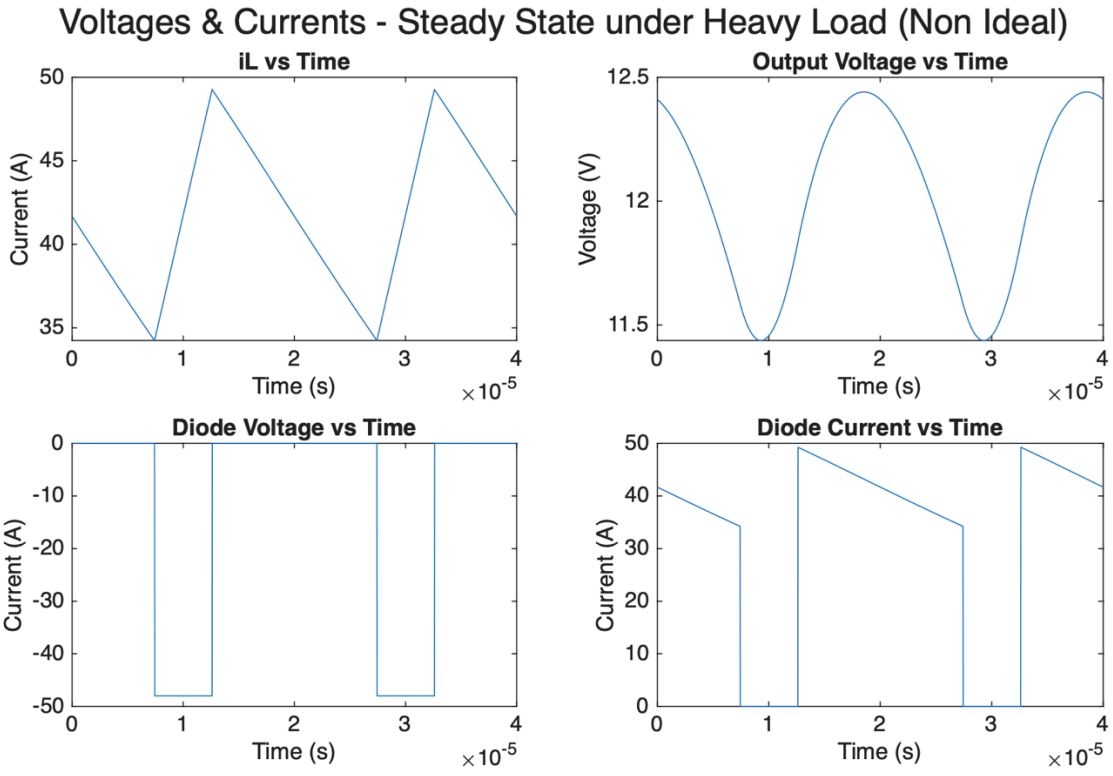
$$= 517.095\ W$$

$$\Rightarrow \text{efficiency} = \frac{500}{517.095} = \boxed{0.967\ \%}$$

a

1f.


Voltages & Currents - Startup under Heavy Load (Non Ideal)
iL vs Time
Current (A)
Time (s)
×10^-4
Output Voltage vs Time
Voltage (V)
Time (s)
×10^-4
Switch Current vs Time
Current (A)
Time (s)
×10^-4
Diode Current vs Time
Current (A)
Time (s)
×10^-4


Voltages & Currents - Steady State under Heavy Load (Non Ideal)
iL vs Time
Current (A)
Time (s)
×10^-5
Output Voltage vs Time
Voltage (V)
Time (s)
×10^-5
Diode Voltage vs Time
Current (A)
Time (s)
×10^-5
Diode Current vs Time
Current (A)
Time (s)
×10^-5

The average value for the Load Current is 41.7175A.
The average value for the Load Voltage is 12.0146 V.
The average value for the Switch1 Current is 10.8254A.

From these values, it can be determined that the power outputted is 500.61W (41.7175A * 12.0146V), and the input power is 519.62W (10.8254A * 48V). Thus, demonstrating an efficiency of 0.963 or 96.3% which is very similar to the analytically determined value of 96.7%. These graphs show the converter meets spec as the Load Voltage is within 12.5 to 11.5V, and it oscillates around 12.01V.

Given the graphs and average power, the proposed Buck Converter Design can be determined to be theoretically successful as it meets all the specs provided by the guideline.

1ci. swch.m

```
function [state] = swch(duty, time)


%make a triangle wave from 0-1

a0 = 0.5; %dc offset
amplitude = 0.5; %
freq = 50000; %period of waveform
w = 2 * pi  * freq;


%f(x) = 8A/(pi^2) * sum [1/k^2 * cos(kwt)]


k = 1:2:50;
triangle_general = (8 / pi^2) * sum((1 ./(k.^2)) .* cos(k * w * time))
triangle = amplitude * triangle_general + a0

if triangle < duty
    state = 1; %on

else
    state = 0; % off

end
```

1cii. bck.m

```
while t(k) < tend

    state = swch(duty,t(k));

   %inductor current
    vs = 48;



    %{

    %IDEAL
    origval = [iL(k); vLoad(k)];
    mult = [(timestep * state * vS / inductance); 0];
    A = [1,(timestep / inductance); (- timestep / capacitance), (1 + timestep / (rLoad
* capacitance))];
    newval = inv(A) * (origval + mult);
    iL(k + 1) = newval(1);

    %}

    %NONIDEAL
    origval = [iL(k) / (1 + timestep * rTon/ inductance) ; vLoad(k)];
    mult = [(timestep * state * vS / inductance)/ (1 + timestep * rTon/ inductance);
0];
    A = [1,(timestep / inductance); (- timestep / capacitance), (1 + timestep / (rLoad
* capacitance))];
    newval = inv(A) * (origval + mult);
    iL(k + 1) = newval(1);

    if(iL(k + 1) < 0)
        iL(k + 1) = 0;
    end

    vLoad(k + 1) = newval(2);


    %cap
    vC(k + 1) = vLoad(k + 1);
    iC(k + 1) = iL(k + 1) - vLoad(k+1)/rLoad;

    %inductor voltage
    vL(k + 1) = (vs - vLoad(k + 1)) * state - (vLoad(k + 1)) * (1 - state);


    %switch 1
    vsw1(k + 1) = vs *  (1 - state);
    isw1(k + 1) = (state) * iL(k + 1);

    %switch 2
    vsw2(k + 1) = -vs * ( state );
    isw2(k + 1) = iL(k + 1) * (1 - state);

    %update time step (delta t)
    t(k) = t(k) + timestep;
    k = k + 1;

end
```

1ciii. bckprc.m

```
function bckprc


    capacitance = 0.000036232;
    inductance = 0.00001242;
    fsw = 50000;
    period = 1/fsw;
    timestep = period / 1000;
    rLoad = 0.288;
    tend = 0.0004;
    t = 0:timestep:tend;

    iL = zeros(size(t));
    vC = zeros(size(t));
    vL = zeros(size(t));
    iC = zeros(size(t));
    vLoad = zeros(size(t));
    isw1 = zeros(size(t));
    isw2 = zeros(size(t));
    vsw1 = zeros(size(t));
    vsw2 = zeros(size(t));
    i2 = zeros(size(t));
    i1 = zeros(size(t));

    rTon = 0.01;
    rDon = 0.01;

    vsw2(1) = -48;
    vS = 48;

    k = 1;
    duty = 0.2586; %.25 for ideal

    vC(1) = 0;
    vLoad(1) = 0;
    iL(1) = 0;

    bck;

    % for getting SS
    removed = round((tend - 2 * period) / timestep)

    iL(1:removed ) = [];
    vsw2(1:removed ) = [];
    isw2(1:removed ) = [];
    vLoad(1:removed ) = [];

    timeAxis = 0:timestep:(tend - removed * timestep);
    %timeAxis = 0:timestep:tend;


    subplot(2, 2, 1); % Selects the first position (top-left)
    plot(timeAxis, iL);
    title('iL vs Time');
    xlabel("Time (s)");
    ylabel("Current (A)");

    subplot(2, 2, 2); % Selects the first position (top-left)
```

```
    plot(timeAxis, vLoad);
    title('Output Voltage vs Time');
    xlabel("Time (s)");
    ylabel("Voltage (V)");

    subplot(2, 2, 3); % Selects the first position (top-left)
    plot(timeAxis, vsw2);
    title('Diode Voltage vs Time');
    xlabel("Time (s)");
    ylabel("Current (A)");


    subplot(2, 2, 4); % Selects the first position (top-left)
    plot(timeAxis, isw2);
    title('Diode Current vs Time');
    xlabel("Time (s)");
    ylabel("Current (A)");

    sgtitle('Voltages & Currents - Steady State under Light Load');

    %averg(iL, period, timestep);
    %averg(vLoad, period, timestep);
    %averg(isw1, period, timestep);



    %{
    subplot(2, 4, 7); % Selects the first position (top-left)
    plot(timeAxis, vsw2);
    title('vsw2 vs Time');
    xlabel("Time (s)");
    ylabel("Voltage (V)");

subplot(2, 4, 4); % Selects the first position (top-left)
    plot(timeAxis, iC);
    title('iC vs Time');
    xlabel("Time (s)");
    ylabel("Current (A)");

 subplot(2, 4, 1); % Selects the first position (top-left)
    plot(timeAxis, vL);
    title('vL vs Time');
    xlabel("Time (s)");
    ylabel("Voltage (V)");

     subplot(2, 4, 5); % Selects the first position (top-left)
    plot(timeAxis, vsw1);
    title('vsw1 vs Time');
    xlabel("Time (s)");
    ylabel("Voltage (V)");
    %}
```

1civ. averg.m

```
function averg(wave, T, dt)

average = 0;
last = length(wave);
period_ints = (T / dt);


for i = (last - period_ints ):1:last - 1;

    average = average + dt * wave(i);

end

average = average / T;
fprintf('average value for waveform is %.4f \n',  average)
```